

Zander Chown

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EDUCATION

- Ph.D. in Computer Science**, *University of Oregon, Eugene, OR* *Sep 2025 – Present*
• Satellite networking research with Prof. Ram Durairajan
- B.A. in Computer Science & Political Science**, *Skidmore College, Saratoga Springs, NY* *Sep 2021 – May 2025*
• Computer Science major GPA: 4.0
• Study abroad: DIS Copenhagen — Spring 2023
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PUBLICATIONS

- **2026** — *SS-Plane: A Heliocentric Satellite Constellation for Demand-Aware Networking*. Under review.
 - **2026** — *Falling Into Line: Scheduling (Re)Deployment and Transition in LEO Satellite Constellations*. Under review.
 - **2024** — *Privacy Norms for Fertility Data in the Roe v. Wade Era*. ACM CCS (Poster). <https://dl.acm.org/doi/pdf/10.1145/3658644.3691406>
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HONORS AND AWARDS

- **Seely Fellowship** — University of Oregon, 2026
 - **Robert and Marcia DeSieno Prize** for Excellence in Computer Science — Skidmore College, 2025
 - **Dean's List** — Skidmore College, Spring 2022 & Fall 2023
 - **Pi Sigma Alpha** — Political Science National Honor Society
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RESEARCH EXPERIENCE

- Graduate Research Assistant**, *Oregon Networking Research Group, University of Oregon* *Sep 2025 – Present*
• Researching the sustainability of satellite constellations and optimized constellation design with Prof. Ram Durairajan and Dr. Chris Misa
• Co-authored *SS-Plane*, a novel satellite constellation design optimized for demand-aware networking (under review, 2026)
• Co-authored *Falling Into Line*, analyzing the scheduling of deployments and transitions in LEO constellations (under review, 2026)
- Undergraduate Student Lead Research**, *Privacy & Security in Digital Health, Skidmore College* *Jul 2022 – May 2025*
• Conducted a comprehensive study on the privacy and security of period-tracking and fertility apps
• Applied the contextual integrity framework to analyze data-protection mechanisms; developed and obtained IRB approval for the protocol
• Designed and ran a multi-survey methodology, surveying several hundred participants and using statistical analysis to evaluate privacy concerns
• Implemented multi-party computation (MPC) techniques to enhance data security, using the Virtual Ideal Functionality Framework to address user concerns
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RESEARCH PRESENTATIONS

- **2025** — *SS-Plane: A Heliocentric Satellite Constellation for Demand-Aware Networking*. UO Department of Computer Science Graduate Research Open House
 - **2024** — *Privacy Norms for Fertility Data in the Roe v. Wade Era*. ACM CCS (Poster)
 - **2023, 2025** — *Securing Mobile Fertility App Data Using Multi-Party Computation*. Skidmore Academic Festival
 - **2022** — *Defining Privacy Norms for Fertility Data*. NY6 Undergraduate Research Conference
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TEACHING

Graduate Teaching Assistant, *University of Oregon**Sep – Dec 2025*

- Taught lab sessions on Unix and C programming to undergraduate students
- Developed exam questions and graded homework and exams

Peer Academic Coach, Computer Science, *Skidmore College**Jan 2023 – May 2025*

- Provided one-on-one and group tutoring sessions for Computer Science students
- Led workshops and study sessions to enhance student learning and foster collaboration

Teaching Assistant, Intro to CS, *Skidmore College**Sep – Dec 2022 & 2024*

- Held office hours and helped students understand course material in a supportive learning environment

PROFESSIONAL DEVELOPMENT

Oregon Programming Languages Summer School (OPLSS), *University of Oregon**June 2026*

- Selected to attend the Oregon Programming Languages Summer School

SKILLS

Programming Languages

C, Python, Zig, Bash, C++, Haskell, Java, R, Swift, Lua, JavaScript, TypeScript, GLSL, WGSL

Technologies & Tools

Git, GitHub, Vim, Xcode, Docker, AWS, Firebase, TensorFlow, NumPy, Pandas, BabylonJS, Make, CMake, Cabal

NOTABLE PROJECTS

Ursus · *Zig*

- UCI-compliant chess engine written in Zig, reaching a rating of ~3300 Elo
- Alpha-beta search with an NNUE evaluation trained from self-play data
- Implements advanced search techniques — null-move pruning, futility pruning, late-move reductions, singular extensions, and more — all verified with SPRT and unit tests
- Source: github.com/zchown/Ursus · plays on Lichess as [@Ursus_bot](#)

Tak · *C, Zig, Haskell, JavaScript, BabylonJS, Python, Accelerate, TensorFlow*

- Implemented the board game Tak across multiple languages and platforms
- Built *Haliax*, a TEI-compliant Tak engine in Zig that plays on [playtak.com](#), using an optimized alpha-beta search with a hand-crafted evaluation function
- Created a web-based 3D UI with BabylonJS and a Haskell game-server manager bridging human and AI players
- Implemented a Monte Carlo Tree Search AI in C with a neural-network policy and value function trained from self-play, compiled to Apple's Accelerate framework for fast on-device inference

Neovim Plugins · *Lua, Vimscript*

- [Continuity.nvim](#) — automatic per-project session management (saves and restores editing sessions); several GitHub stars
- [nvim-ipyb](#) — runs and reports Python results notebook-style for quick iteration

MindMNIST · *Python, NumPy, TensorFlow*

- Classified EEG data using machine-learning techniques
- Cleaned data with Fast Fourier Transform methods (e.g., bandpass filtering)
- Compared convolutional networks against K-means and K-nearest-neighbor classifiers
- Full writeup on [Medium](#)

Weather Wear · *Swift (iOS)*

- Built an iOS app combining weather, alarms, and clothing recommendations
- Integrated weather APIs for real-time data; applied object-oriented design patterns (singleton, observer)
- Collaborated with a team using Git for version control

World Terrain Generation · *BabylonJS, GLSL*

- Built a collaborative 3D graphics project with custom GLSL shaders for dynamic lighting and texturing
- Hosted at [benno-lueders.de](#)